Cyclic Freight Service Design with Cross-Dock Handling

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OR-Bench candidate problem • August 2026

Abstract. Freight carriers design repeating line-haul schedules with a finite fleet of reusable tractors. A feasible plan must route each shipment through time-compatible services and close every tractor rotation within the weekly fleet budget. This benchmark isolates that interaction in an eight-leg periodic network with one cross-dock transfer and two available tractors. The unique optimum operates a direct outbound leg and its return, at a weekly cost of 15. Generated solver programs exhibit two distinct formulation errors. One duplicates destination demand and declares the instance infeasible. Two others undercount tractor use and report 11.5 for a schedule that requires three tractors. Exhaustive enumeration of all 2^8 service designs certifies the optimum and both failure modes.

Keywords. service network design, cross-docking, periodic time-space network, fleet balance, mixed-integer linear programming

Form domain: Supply chain and logistics • Model: MILP • Exact oracle: all 2^8 service-leg designs

1. Business Brief

Business-level description. *A regional freight carrier is redesigning a line-haul schedule that repeats every week. Management must decide which candidate services to operate and how to route one urgent load from its origin to its destination. Every operated service requires an owned tractor. Tractors may wait at terminals, but rentals and unlisted repositioning moves are unavailable. The entire weekly schedule must be covered by at most two tractors. The load may travel directly or transfer through a cross-dock, where unloading, sorting, and reloading take one full time slot. Management wants the least-cost schedule that delivers the load on time and remains executable every week with the available fleet.*

The business request couples three decisions that are often modeled separately: service selection, shipment timing, and reusable-asset deployment. A locally balanced time-space network can still be operationally impossible when its periodic walks consume more tractor time than the fleet provides. The benchmark is designed to expose that gap.

2. From Business Request to Benchmark Instance

We encode the request as one canonical instance, which is the unit evaluated in OR-Bench. The instance is deliberately small enough for complete verification while retaining the two operational mechanisms that create the modeling challenge. Let $N = \{O, H, D\} \times \{0, 1, 2, 3\}$ be the periodic time-space nodes, A the candidate tractor legs, and B the zero-cost one-slot waiting arcs. A return leg reaches its stated terminal in the next weekly cycle. Each candidate service can operate at most once per week.

Table 1. Candidate service legs. Return legs are cargo-ineligible.

Leg	Tail	Head	Cost	Cargo
$F1$	$O0$	$H1$	2	yes
$R1$	$H1$	$O0$ next cycle	2	no
$F2$	$H1$	$D2$	2	yes
$R2$	$D2$	$H1$ next cycle	2	no
$F3$	$H2$	$D3$	10	yes
$R3$	$D3$	$H2$ next cycle	10	no
$F4$	$O0$	$D3$	7.5	yes
$R4$	$D3$	$O0$ next cycle	7.5	no

One indivisible load is released at $O0$ and must reach D no later than slot 3. The one-slot cross-dock operation moves the freight state from $H1$ to $H2$. The two feasible cargo paths are therefore

$$\mathcal{R} = \{r^{\text{dir}} = (F4),\ r^{\text{xdock}} = (F1, F3)\}.$$

The apparently inexpensive pair $(F1, F2)$ is not a valid itinerary. Leg $F1$ reaches the hub in slot 1, the cross-dock operation finishes in slot 2, and leg $F2$ has already departed in slot 1. This timing detail separates a valid service path from a merely connected path.

3. Ground-Truth MILP

Let $y_a \in \{0,1\}$ indicate whether service leg a operates, let $w_b \in \mathbb{Z}_+$ count tractors on waiting arc b, and let $z_r \in \{0,1\}$ select cargo path $r \in \mathcal{R}$. Write d_e for arc duration, $H = 4$ for the weekly horizon, and $M = 2$ for the fleet. The ground-truth MILP is

$$\min \quad \sum_{a\in A} c_a y_a, \tag{1}$$

$$\text{s.t.} \quad \sum_{r\in\mathcal{R}} z_r = 1, \tag{2}$$

$$z_r \le y_a \qquad \forall r \in \mathcal{R},\ a \in r, \tag{3}$$

$$\sum_{a\in A\cap\delta^+(n)} y_a + \sum_{b\in B\cap\delta^+(n)} w_b = \sum_{a\in A\cap\delta^-(n)} y_a + \sum_{b\in B\cap\delta^-(n)} w_b \qquad \forall n \in N, \tag{4}$$

$$\sum_{a\in A} d_a y_a + \sum_{b\in B} d_b w_b \le HM = 8, \tag{5}$$

$$y_a, z_r \in \{0,1\}, \quad w_b \in \mathbb{Z}_+. \tag{6}$$

Equation (2) selects one itinerary, and (3) opens its services. Equation (4) closes the tractor circulation at every periodic node, while (5) limits total tractor-time to $HM = 8$ tractor-slots. Path generation enforces the hub connection rule $\mathrm{dep}(b) \ge \mathrm{arr}(a) + h_H$, where $h_H = 1$.

4. Optimal Design and Diagnostic Evidence

The direct itinerary operates $F4$ and $R4$. These legs form a four-slot rotation covered by one tractor at a weekly cost of 15. The cross-dock itinerary requires $F1$, a hub wait, $F3$, and $R4$, at a cost of 19.5. The direct rotation is therefore the unique optimum. Exhaustive evaluation also measures the effect of omitting handling time, tractor circulation, or the fleet budget.

Table 2. Correct model and four diagnostic ablations.

Model	Cargo path	Operated services	Objective
Ground truth	$F4$	$F4$, $R4$ (1 tractor)	**15**
No handling time	$F1, F2$	$F1, F2, R1, R2$ (2)	8
No fleet limit	$F4$	$F4, R1, R2$ (3)	11.5
No tractor constraints	$F4$	$F4$	7.5
No handling or tractors	$F1, F2$	$F1, F2$	4

Failure certificate. A low-reasoning GPT-5.4 program dated 2026-03-05 returns INFEASIBLE. Its shipment balance assigns one unit of supply at $O0$ but one unit of demand at both $D2$ and $D3$. It therefore requires two deliveries for one load. The explicit rotation $F4, R4$ is a feasible certificate with objective 15.

A second generated Gurobi model reports 11.5 with services $F4, R1, R2$. Its periodic walk uses $F4$, three waits at D, $R2$, and $R1$, for $3+3+3+3 = 12$ tractor-slots. Repeating those departures every four slots requires $12/4 = 3$ staggered tractors, although only two are available. The missing fleet-time restriction understates the optimum by 3.5, or 23.33%.

Two independent GPT-5-mini runs dated 2025-08-07 reproduce the 11.5 error. One collapses a winding periodic walk into one named-tractor circulation, and the other undercounts cross-week arcs. High-reasoning GPT-5.4, GPT-5.5, and GPT-5.6 Luna runs recover the correct objective. The instance therefore separates shipment-balance errors from periodic fleet-accounting errors.

5. Benchmark Package and Sources

The public package provides one fully specified benchmark instance, including the business brief, precise statement, original CSV data, and a Gurobi solver that verifies the ground truth and four ablations. The original statement and data are CC BY 4.0, and the solver is MIT licensed. Package: `https://github.com/Zhengzhong-You/OR-Bench`.

The periodic time-space and vehicle-balance structure follows Gao, Jin, and Diao [1]. The one-slot transfer semantics is motivated by Google's Shipping Network Design documentation [2].

References

[1] A. Gao, X. Jin, and X. Diao. 2022. An iterative backbone algorithm for service network design problems. *Processes* 10(7):1373. doi:10.3390/pr10071373.

[2] Google for Developers. 2024. Shipping Network Design, Operations Research API. Shipping Network Design documentation.